book, I have depicted a black hole's singularity (Figure 26.9, for example).

Since Wheeler's era, mathematical calculations with Einstein's laws have taught us that these pointy singularities are unstable. To create them inside a black hole requires precise tuning. When perturbed ever so slightly, for example by something falling in, they change enormously. Change into what?

Three Russian physicists—Vladimir Belinsky, Isaac Khalatnikov, and Eugene Lifshitz—used long, complicated calculations to guess the answer, in 1971. And in the 2000s, when computer simulations became sufficiently advanced, their guess was confirmed by David Garfinkle at Oakland University. The resulting, stable singularities now carry the name BKL in honor of Belinsky, Khalatnikov, and Lifshitz.

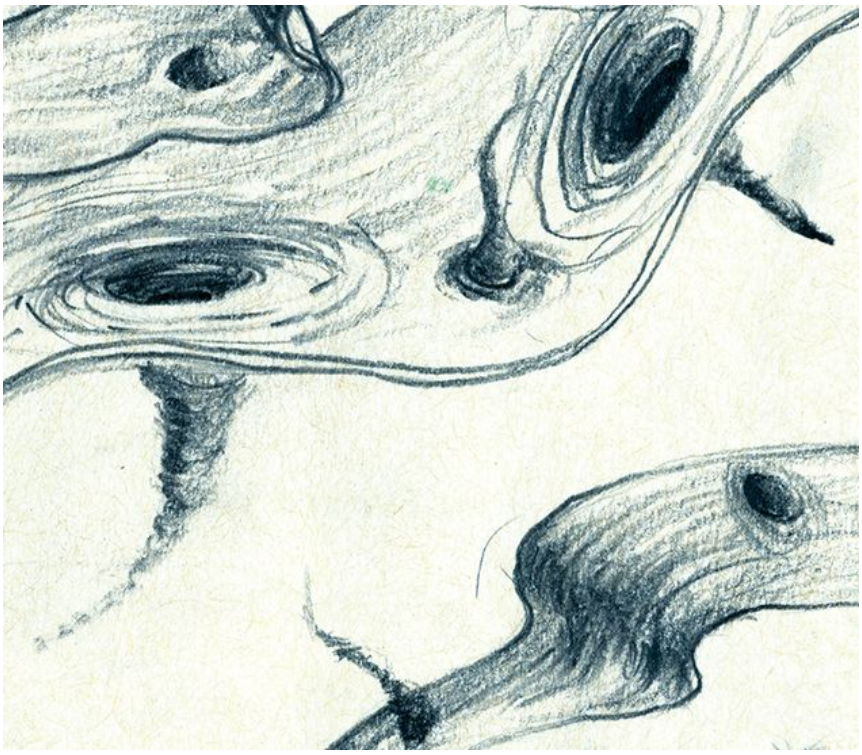


Fig. 26.9. Lia Halloran's fanciful drawing of several black holes with singularities at their pointy tips. *[A segment out of Fig. 4.5.]*